Viva sheet: Delhi AQI predictive analytics

One line per technique: what it does, and what it showed in this project

The project in five sentences

1. Delhi, 1,826 days (2015-2019), nine pollutants and AQI from the Kaggle CPCB file; 2020 held out as a lockdown test.
2. PM2.5 and PM10 drive AQI (r = 0.88; β 0.43 and 0.38); 68% of days are Poor or worse.
3. Polynomial regression predicts AQI with R^2 0.885; logistic regression predicts Poor days with AUC 0.981.
4. A trend + month regression forecasts monthly AQI 6 months ahead with 11.8% MAPE, 27% better than “same month last year”.
5. Control charts flag the 2016 smog, the post-2017 improvement and the 2020 lockdown (4-7σ below the limits).

The caveat to say first: AQI is computed from the pollutants, so the same-day models partly recover the formula; forecasting is the genuinely predictive part.

Data wrangling

Technique	What it does	What it showed
Schema and duplicate check	confirms columns, removes repeated rows	16 expected columns, 0 duplicates, complete daily calendar
Hidden-missing check	finds impossible values coded as numbers	CO = 0 for 84 days (Jun-Aug 2017): sensor outage, recoded as missing
Masked-value experiment	hides known values and scores imputation methods	interpolation best for short gaps (nRMSE 0.48), MICE for long gaps (1.08)
Linear interpolation	straight line between neighbouring days	best for 1-7 day gaps because AQI is persistent
MICE (iterative imputation)	predicts a missing pollutant from the others + season	best for long PM2.5/PM10/NO2 gaps
IQR rule	flags points beyond Q1 - 1.5 IQR / Q3 + 1.5 IQR	flags 3-10% of values, mostly genuine winter smog
Hampel filter	flags spikes vs a rolling median	isolated spikes flagged, kept
Winsorisation	caps extreme values at percentiles	0.5/99.5 percentile caps, about 20 values per pollutant; AQI untouched
Data contract	automated checks on the cleaned file	20/20 checks pass

Exploratory analysis

Technique	What it does	What it showed
Frequency table / bar / pie	counts days per AQI band	1.3% Good, 12.9% Severe; winter has no Good day
Mean, median, mode	central tendency	AQI mean 266.3 ≈ median 266.0; PM2.5 mean 120 > median 98
SD, variance, CV	spread	PM2.5 CV 68%; AQI SD 120
Skewness	asymmetry (3rd moment)	PM2.5 1.39 (right-skewed); AQI only 0.26
Kurtosis (excess)	tail weight (4th moment)	PM2.5 2.12 heavy-tailed; AQI -0.35 slightly flat
Normality tests (D'Agostino, JB)	test for a normal distribution	all reject; check residuals, not raw variables
Kruskal-Wallis / ANOVA	compare seasons	$H = 930$, $\varepsilon^2 = 0.51$: season explains half the variation
Pearson / Spearman correlation	linear / rank association	PM2.5 0.88/0.92; CO 0.27/0.54 (monotonic, non-linear)
Fisher z interval	CI for a correlation	PM2.5 r 0.874-0.894
t confidence interval	CI for a mean	260.8-271.8, too narrow because days are dependent
Effective sample size	$n(1-\rho)/(1+\rho)$ for autocorrelated data	1,826 days ≈ 98 independent; CI 242-290
Wilson / block bootstrap CI	CI for a proportion	Poor share 68.0%; honest CI 59-76%

Regression

Technique	What it does	What it showed
Simple linear regression	AQI = β0 + β1·PM2.5	AQI = 109.3 + 1.30·PM2.5, R^2 0.783
t and F tests	significance of slope / model	t = 72.6, F = 5,269 = t^2; p < 0.001
Confidence vs prediction interval	uncertainty of mean vs single day	at PM2.5 = 100: CI 237-243, PI 130-349
Residual analysis	checks model assumptions	near-normal; DW 0.69 shows strong autocorrelation
Multiple regression	many predictors at once	R^2 0.881; PM2.5 and PM10 dominate
Standardised (beta) coefficients	compares predictors in SD units	PM2.5 0.43, PM10 0.38, CO 0.13
Dummy variables	categorical predictor (season)	monsoon 32.6 points below winter at equal pollution
VIF	multicollinearity	all < 6 (moderate)
Breusch-Pagan	heteroscedasticity	p ≈ 1e-15: variance grows with AQI
HC3 robust SE	inference despite heteroscedasticity	no verdict changed
Durbin-Watson	autocorrelation of errors	1.10 in date order: positive autocorrelation
Cook's distance / leverage	influential points	max 0.05 (< 1); found an impossible PM2.5 > PM10 day
Log transformation	stabilise variance	made things worse (RMSE 70 vs 41), rejected
Backward elimination by AIC	model building	dropped SO2, NO2, NO
Polynomial term	curvature	$PM2.5^2$ significant; effect falls from 1.22 to 0.47 per µg/m^3
Partial F-test	tests a group of terms	season jointly significant (F = 18.6)
10-fold CV / 2019 hold-out	honest generalisation	polynomial best: CV RMSE 40.2; 2019 R^2 0.905

Logistic and multinomial regression

Technique	What it does	What it showed
Logistic function	maps a score to a probability, 1/(1+e^-η)	median monsoon day P(Poor) = 0.025
Likelihood-ratio test / pseudo R^2	overall fit	p < 0.001; McFadden 0.741
Wald test	significance of each coefficient, $(β/SE)^2$	PM2.5, PM10, NOx, CO, O3, NO2 significant
Odds ratio	e^β, multiplicative change in odds	+10 µg/m^3 PM2.5 → odds × 1.69
Classification table	confusion matrix at a threshold	11 missed Poor days, 11 false alarms, accuracy 0.940
ROC curve / AUC	ranking quality across thresholds	AUC 0.981
Gini coefficient	2·AUC - 1	0.961
Hosmer-Lemeshow	calibration goodness of fit	p = 0.73 (train), 0.60 (test): good fit
Cost-based threshold	trades misses vs false alarms	3:1 cost → threshold 0.35, misses 11 → 7
Quasi-separation	perfect prediction breaks MLE	why season is left out of the multinomial model
Multinomial logit	K-1 logit equations vs a baseline	accuracy 0.902, macro F1 0.854, AUC 0.973
Relative risk ratio	e^β in the multinomial model	PM2.5 decisive; PM10 matters only for High vs Low

Decision trees

Technique	What it does	What it showed
CART classification tree	binary splits minimising Gini impurity	root split PM2.5 ≤ 76.3 (near the official 90 breakpoint)
Cross-validated tuning	chooses depth and leaf size	depth 8, ≥ 20 days per leaf
Cost-complexity pruning	removes weak branches (α)	23 → 20 leaves, same accuracy
Feature importance	total impurity reduction	PM2.5 0.76, PM10 0.10
CART regression tree	splits minimising squared error	R^2 0.886, RMSE 40.3, lowest MAE
CHAID idea (chi-square)	splits on the most significant χ^2	season vs AQI group χ^2 779.6, Cramér's V 0.46
Random forest (benchmark)	many averaged trees	AUC 0.982, no better than logistic
Unstructured data	text and images turned into features	fire counts and festival dates are the natural extension

Forecasting and time series

Technique	What it does	What it showed
STL / classical decomposition	trend + seasonal + remainder	seasonal strength 0.85; Nov +143, Jul -139
ADF / KPSS	unit-root / stationarity tests	level non-stationary; differenced series stationary
ACF / PACF	identify AR and MA orders	AR(1) plus seasonal MA(1) at lag 12
Moving average forecast	mean of the last k values	MA(3) fails on a seasonal series (MAPE 62%)
Simple exponential smoothing	geometrically weighted average	α = 0.995 → same as naive; fails
Holt-Winters	level + trend + seasonal smoothing	2nd best, 18% skill
ARIMA(p,d,q)	Box-Jenkins without seasonality	(2,1,2): MAPE 46%, poor
SARIMA	adds seasonal terms	$(1,0,0)(0,1,1)_{12}$; seasonal MA ≈ -1 → deterministic seasonality
Ljung-Box	residual autocorrelation	SARIMA residuals white (p = 0.61)
Causal regression forecast	trend + month dummies (+ lagged pollutants)	**best**: rolling MAPE 11.8%, trend -19 AQI per year
GLS with AR(1) errors	corrects autocorrelated errors	ρ = 0.43; honest SEs, no accuracy gain
VAR	each series on lags of all series	VAR(3); NO2 and PM10 Granger-cause AQI
Granger causality	does X's past help predict Y?	NO2 → AQI p = 0.001; PM2.5 → AQI no (already in AQI's past)
Impulse response / FEVD	shock propagation / variance shares	PM10 explains 34% of AQI 6-month forecast variance
RMSE / MAPE	forecast accuracy	test RMSE 29.0, MAPE 13.9%
Rolling-origin evaluation	time-series cross-validation	19 origins; robust model ranking
Prediction interval	range for future values	95% PI ±85; Jan-Feb 2020 inside
Counterfactual forecast	what would have happened	Apr-Jun 2020 actual 36% below forecast = lockdown effect
ARMA errors, daily	next-day forecast	RMSE 50.2 vs persistence 54.0

Control charts

Technique	What it does	What it showed
I-MR chart	individuals with $\sigma = \bar{MR}/1.128$	raw: mostly flags the season (14/60)
Seasonal adjustment	removes calendar effect	real signals: Oct-Nov 2016 smog, downward shift after 2017
Western Electric rules	1 beyond 3σ; 2/3 beyond 2σ; 4/5 beyond 1σ; 8 on one side	runs reveal a level shift
$\bar{X}$-S chart	subgroup means and SDs	45/60 outside: months are not rational subgroups
p-chart	proportion defective, variable n	36/60 outside: over-dispersion (χ^2 1,015 on 59 df)
Laney p′ chart	widens p limits by σz	σz 1.97; monsoon months remain low signals
Phase II monitoring	apply fixed limits to new data	Mar-Jun 2020 4.3-6.8σ below LCL: COVID-19 lockdown

Likely viva questions

- **Why not delete outliers?** They are real smog days, the very events studied; only sensor extremes were capped.
- **Why is the t-interval wrong?** Daily AQI is autocorrelated (r = 0.90), so the effective sample size is about 98.
- **Why does NH3 have a negative coefficient?** AQI is a *maximum* of sub-indices and NH3 correlates with PM2.5; it is not causal.
- **Why did the simplest forecasting model win?** The seasonality is deterministic (seasonal MA ≈ -1), so fixed month effects plus a trend capture it with the fewest parameters; SARIMA and VAR spend parameters on noise.
- **Why keep 2020 out?** The lockdown is a structural break; it is used as a natural experiment instead.
- **What would you add next?** Weather (wind, inversion height, rain) and satellite fire counts for the Oct-Nov peak.